



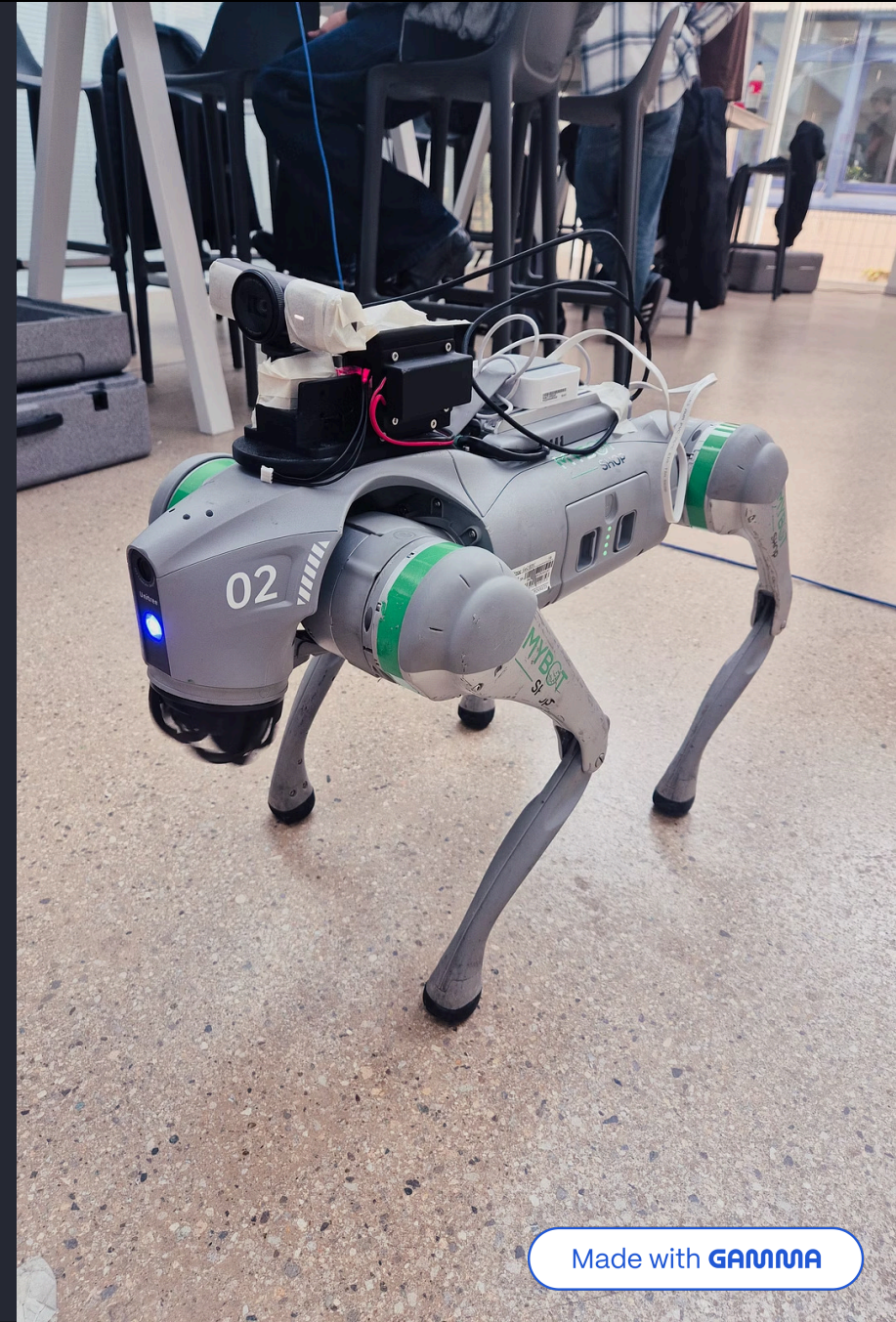
ArgOS

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The Problem

Robot dogs are here, what's missing?

- ☐ natural, intuitive control
- ☐ understanding of the world around them
- ☐ easy extensibility



The Solution: ArgOS

a platform built on top of DimOS that lets you control any robot dog using natural language



Sees

Visual perception and object segmentation



Remembers

Builds spatial memory of detected objects



Understands

Interprets natural language commands (voice + text)



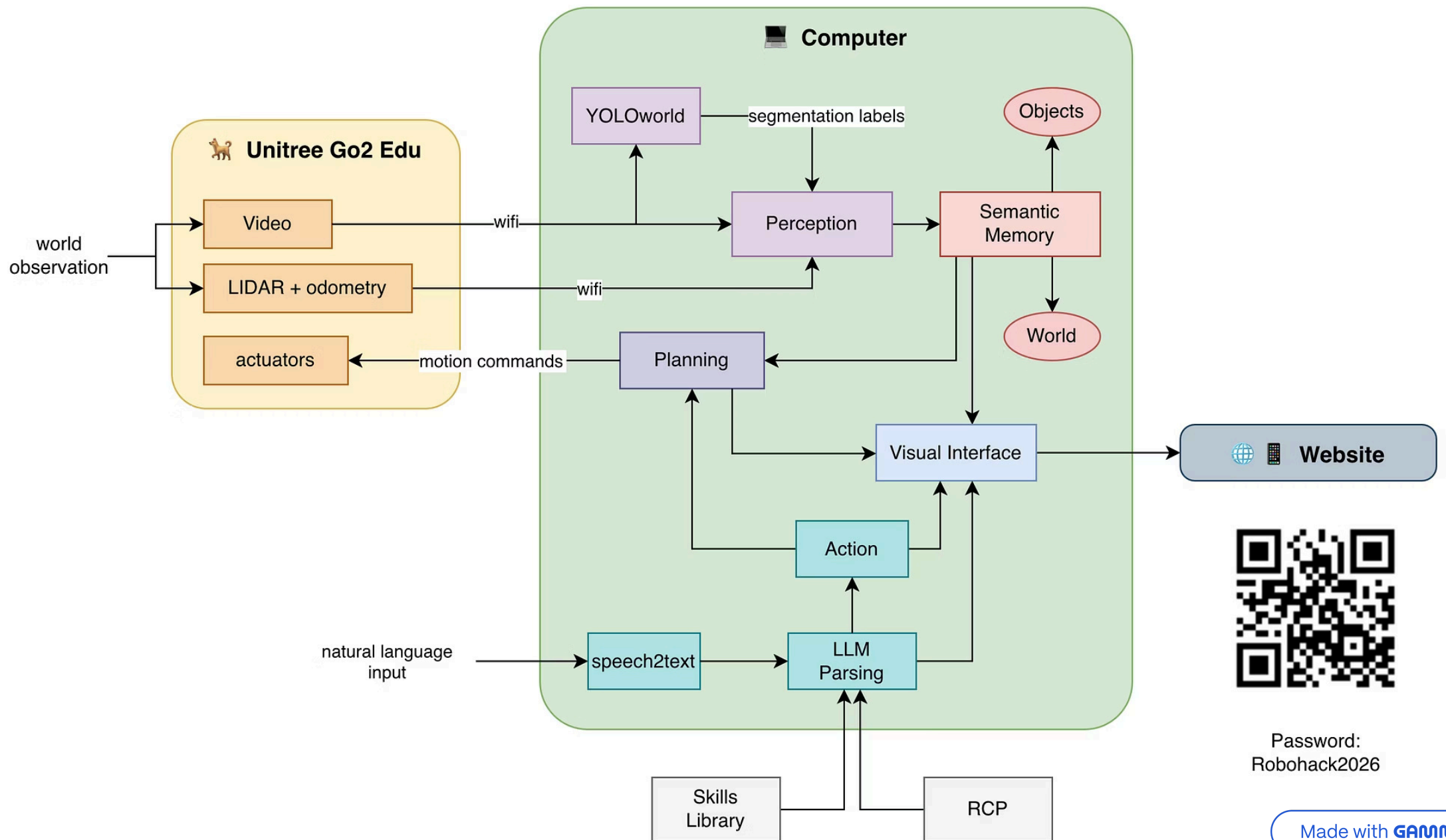
Navigates autonomously

SLAM, obstacle avoidance, path planning



Works anywhere

Control and visualize using your browser



Password:
Robohack2026

Key Features



Natural language control

voice + text via browser



Extensible MCP tool system

add new skills easily



Real-time object segmentation

20 FPS on-board, 90+ FPS off-board



Advanced navigation

build maps in real-time, plan paths, avoid obstacles



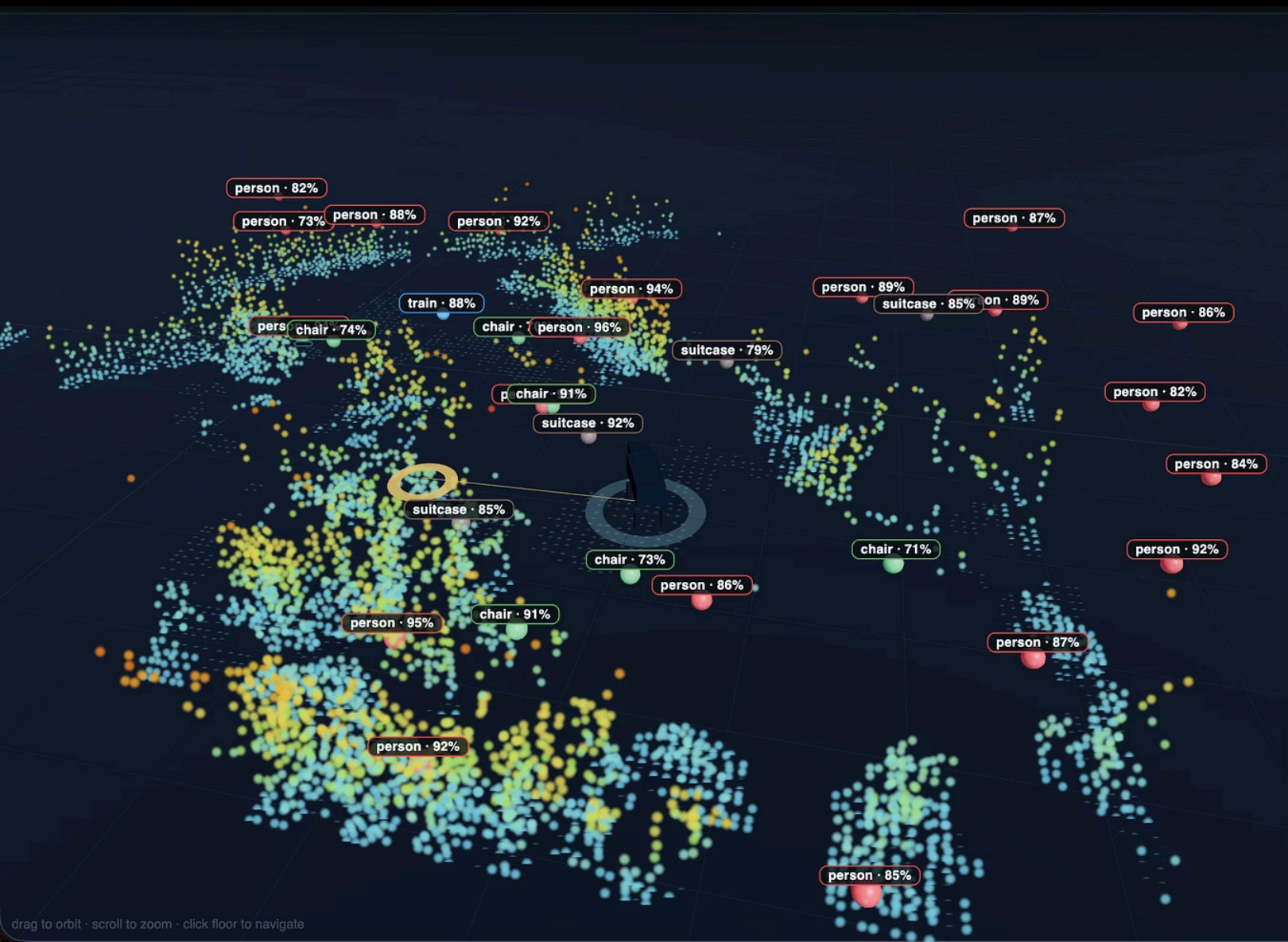
Semantic spatial memory

remembers where objects are in 3D



Agentic autonomy

search, and navigate to targets with a single command



drag to orbit · scroll to zoom · click floor to navigate

ASK

AGENT

Where is the chair?

Ask

CAMERA



OBJECTS IN SEMANTIC MEMORY

Label	Pos (m)	Conf
chair	(3.6, -3.7)	91%
person	(2.9, -3.9)	95%
person	(3.5, -4.7)	92%
person	(3.3, -6.5)	86%

Use Cases

Security patrol

Inventory/inspection

Accessibility assistance

Delivery

Household tasks

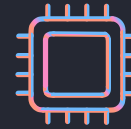
Future Extensions



Open-vocabulary detection
(Grounded SAM)



multi-robot coordination



fully on-board processing



hardware extensions (grippers,
trays, baskets)



person recognition



Gaussian splatting viz



ArgOS

we're teaching old dogs new tricks

Challenges & Learnings

1

Onboard processor cannot access

Solved with USB camera + HTTP bridge.

2

CycloneDDS version conflicts

Encountered issues with different versions.

3

Built visual eviction for stale objects

Implemented a system for better object management.

Live Demo

Experience the robot's advanced capabilities through a series of interactive demonstrations.



People Count

The robot accurately counts and responds to the query, "How many people do you see?"



Object Navigation

The robot efficiently navigates to locate and identify a specific object, demonstrated by "Find the chair."



Real-time Segmentation

Witness real-time object segmentation overlaid directly onto the camera feed, enhancing visual understanding.



Immediate Halt

The robot executes an immediate and precise halt command upon receiving "Stop."



Scene Description

The robot provides a detailed verbal description of its surroundings in response to "What do you see?".